



Prognosis of juvenile myoclonic epilepsy is related to endophenotypes

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ABSTRACT

Purpose: To investigate if phenotypic variations have prognostic implications in juvenile myoclonic epilepsy (JME).

Methods: Sixty-five consecutive JME patients had video-EEG recording and were followed for at least three years. Reflex traits were defined as seizures and/or EEG discharges induction by eye-closure, photic stimulation, language, praxis or calculation. Patients had psychiatric evaluation and answered to STAI (State-Trait Anxiety Inventory). Seizure control was classified according to Prasad et al. (2003).²⁴ Patients were divided into Group 1: good seizure control and Group 2: moderate or poor seizure control. Video-EEG/EEG evaluation was repeated in 21 patients.

Results: Forty of 65 (61.5%) patients reached good seizure control, 25 (38.5%) of whom became seizure free. Group 2 patients had longer epilepsy duration (13.9 ± 9.0 vs. 8.7 ± 8.2 ; $p = 0.019$); higher prevalence of the combination of all three seizure types (72.0% vs. 30.0%; $p = 0.003$); discharges in baseline EEG (56.0% vs. 22.5%; $p = 0.008$); seizure recording (68% vs. 20%; $p < 0.001$) and sensitivity to praxis (63.6% vs. 29.6%; $p = 0.023$). Compared to seizure-free patients, those with persistent seizures presented younger age at epilepsy onset (12.6 ± 3.33 years vs. 15.4 ± 5.47 years; $p = 0.015$); higher prevalence of personality disorders (25% vs. 4%; $p = 0.029$); higher scores in STAI-T (45.9 ± 11.31 vs. 36.6 ± 11.43 ; $p = 0.011$) and higher incidence of sensitivity to praxis (58.6% vs. 25.0%; $p = 0.04$) and to language (53.8 vs. 16.7%; $p = 0.026$) tasks. Repetition of EEG/video-EEG revealed a parallel evolution of reflex traits disappearance and seizure control.

Discussion: Clinical features and reflex traits have prognosis implications in JME.

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1. Introduction

Juvenile myoclonic epilepsy (JME) is the most common idiopathic generalized epilepsy (IGE) in adults, comprising up to 10% of all epilepsy patients and up to 26% of IGE.¹ It is characterized by myoclonic jerks in all cases, generalized tonic-clonic seizures (GTCS) in about 90% and absences in one third of these individuals.²

Seizures usually begin between 12 and 18 years of age and important clues to diagnosis include the influence of precipitant factors traditionally represented by sleep deprivation, stress, menstruation, fatigue, and alcohol consumption.² Besides these factors, seizures in JME can be triggered by cortical stimuli which can be simple as intermittent photic stimulation (IPS)³ and eye closure⁴ or complex as superior cortical functions. Among the latter we can distinguish language tasks as reading and speaking⁵ and activities such as calculation, writing, playing cards or chess, construction or drawing, which involve decision-making, complex

spatial manipulation and response by the use of a part of the body, defined as praxis-induction.^{6,7}

JME has been considered as generally responsive to adequate treatment, presenting rates of pharmacoresistance of around 10–20%.^{8–10} Some data open questions regarding its benign nature reporting seizure relapse in up to 70% of patients after at least one year of seizure control.¹¹ In a more recent population based cohort of 23 patients with a very long follow-up of 25 years, 11 (47.8%) were out of medication at the end, in spite of the fact that three continue presenting myoclonia and two, rare seizures. The rate of seizure freedom for all seizure types in this series was of only 17%. Yet 74% of the patients had at least one major unfavorable social outcome as failure to complete high school, unplanned pregnancy, depression, unemployment or living alone.¹²

Until now, there are insufficient data to allow identification of the factors indicating long-term prognosis in JME. Among the potential prognostic factors, non-classical clinical presentations, as childhood absence epilepsy evolving to JME,¹³ focal EEG abnormalities^{14–16} and psychiatric comorbidities, more commonly represented by anxiety, mood and cluster B personality disorders,^{9,17} have been recognized as factors related to difficulty in achieving seizure control.

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Finally, despite the fact that unfavorable seizure control in JME patients with reflex traits related to higher cortical functions as language and praxis has increasingly been recognized since early 90s,¹⁴ its importance on prognosis has only been reported in two series of patients and a few case reports.^{6,14,18,19}

The purpose of this study was to investigate the influence of each feature mentioned above on response to treatment in a prospective series of JME patients with long-term follow-up. As it done in a tertiary center the main objective was not to evaluate the incidence of pharmacoresistance but to evaluate the differences among responders and difficult to treat patients.

2. Methods

2.1. Clinical evaluation

All patients were followed up in the outpatient clinic of a tertiary center (Epilepsy Section, Department of Neurology and Neurosurgery, Universidade Federal de São Paulo, São Paulo, Brazil). Inclusion criteria were: (a) unequivocal diagnosis of JME based on criteria defined in the revised Classification of Epilepsies and Epileptic Syndromes²⁰; (b) age over 12 years; (c) no evidence of neurological or intellectual deficits and (d) more than 4 years of formal education.

Since different electroclinical subtypes of JME might influence prognosis, patients were subdivided in four groups: classic JME, childhood absence epilepsy persisting and evolving into JME, JME with adolescent onset pyknoleptic absence and JME with astatic seizures.¹³

Patients also had psychiatric interviews as part of their clinical characterization by the same psychiatrist (G.M.A.F.) using two structured questionnaires: the Schedule Clinical Interview for DSM-IV, Axis I (SCID-I) and Axis II (SCID-II), consisting in psychiatric scales based on DSM-IV (*Diagnostic and Statistical Manual of Mental Disorders*, third edition, revised). Axis I refers to the evaluation of most psychiatric diseases (e.g. mood, anxiety and psychotic disorders), while Axis II refers to the evaluation of all types of personality disorders. For those younger than 18 the Brazilian version of the Schedule for Affective Disorders and Schizophrenia for School-Aged Children (K-SADS-PL) was applied. State-Trait Anxiety Inventory (STAI), composed of two scales intended to measure state (STAI-S) and trait (STAI-T) anxiety components was used in order to measure anxiety severity.²¹

2.2. Video-EEG

Consecutive patients who met clinical and EEG criteria for JME diagnosis were submitted to a video-EEG monitoring for a 4–6 h period, comprehending a research protocol approved by the Ethics Committee of Universidade Federal de São Paulo, after informed consent was obtained from all participants. Clinical signs of antiepileptic drug (AED) intoxication, occurrence of a GTCS and use of intravenous AED within the last 72 h were exclusion criteria as well as great difficulty in performing any task independent of schooling.

Video-EEG was recorded on a 32-channel digital equipment (Biologic[®], software Ceegraph[®]) using the 10–20 International Electrode System, in addition to perioral and deltoid electrodes. Medications were maintained in all treated patients. After having slept for at least 6 h, all patients had 30 min of awake EEG recording starting at 7 a.m. After this baseline record, patients performed habitual methods of activation as eye-closure, hyperventilation, IPS, sleep and neuropsychological tasks represented by reading silently and aloud, talking, writing, mental and written calculations, drawing and spatial construction puzzles. The

protocol and its analysis were based on those reported by Matsuoka et al.⁷ and Mayer.²²

Reflex traits were characterized by seizure induction and/or EEG discharges increasing by activation methods in video-EEG and are described elsewhere.²³

EEG tracings were also analyzed regarding the presence of focal abnormalities such as focal slow-wave bursts and focal epileptic discharges; amplitude asymmetry ($\geq 50\%$) of generalized discharges; asymmetry of response to IPS and focal onset of generalized discharges.¹⁶ Bilateral anterior or posterior discharges were not considered focal abnormalities.

After a period of 2–6 years (mean 4.4), 16 patients who had presented reflex traits in the first evaluation had a second video-EEG recording and five, who had not expressed neuropsychological induced traits had data obtained from routine EEG with eye-closure and IPS.

2.3. Follow-up

The various seizure types, possible seizure triggers, AED therapy, treatment adherence and adverse effects were observed as well as routine blood tests, AED level determination, and routine EEGs. Fifty-eight patients were on AEDs when enrolled in the study.

Patients were treated by adjustment in lifestyle being oriented to avoid precipitant factors as sleep deprivation and alcohol consumption. They received sodium valproate (VPA), as the first choice drug, in mono or polytherapy and other AED considered reasonably effective in IGE and JME treatment such as topiramate (TPM), lamotrigine (LTG), phenobarbital (PB) and benzodiazepines. Doses and AEDs were chosen according to clinical response and adverse effects.

Estimation of the frequency of seizures was based on the review of seizure calendars and specific questioning of the patient and family members. Evaluation of seizure control was done according to Prasad et al. [24], as follows: GTCS, good (<1 seizure per year), moderate (1–4 seizures per year), or poor (>4 seizures per year); myoclonia, good (<5 single seizures or clusters per month, rare seizures, or occasional seizures), moderate (5–14 single seizures or clusters per month, several seizures, or frequent seizures), or poor (≥ 15 single seizures or clusters per month or daily seizures); and absence, good (<5 seizures per month, rare seizures, or occasional seizures), moderate (5–14 seizures per month, several seizures, or frequent seizures), or poor (≥ 15 seizures per month, frequent seizures, or daily seizures).

In a first analysis patients were followed for at least three years and classified into two groups, Group 1: good seizure control for all seizure types in the last two years and Group 2: moderate or poor seizure control. Seizure-free patients were included in the good seizure control group. Since the main goal of epilepsy treatment is to get patients seizure-free, in a second analysis we compared those completely seizure-free for at least the last two years to those with any degree of persistence of seizures.

Statistical analysis was performed using computer softwares SPSS for Windows release 10.0.1 standard version; STATISTICA version 7, StatSoft, Inc and STATA 8, Data Analysis and Statistical Software, Stata Corporation. Fisher's exact test for categorical and Student's *t*-test for continuous variables were applied for group comparisons. A *p* value of <0.05 was considered statistically significant.

3. Results

3.1. Clinical features

Among 76 enrolled patients we kept following 65 (85.5%). Six lost follow-up, two had non adherence to treatment, one had

Table 1
Demographic and clinical data.

	JME (n = 65)
Gender male	32 (49.2%)
Age (years)	24.40 ± 7.28
Manual dominance right handed	58 (89.2%)
Age at epilepsy onset (years)	13.70 ± 4.48
Duration of epilepsy (years)	10.73 ± 8.73
Seizure types	
Myoclonia	65 (100.0%)
Generalized tonic-clonic	62 (95.3%)
Absences	31 (47.7%)
Family history	46 (70.8%)
JME subtype	
Classic	48 (73.85%)
CAE evolving to JME	7 (10.77%)
With adolescent onset absence	10 (15.38%)
With astatic seizures	0 (0.0%)
Psychiatric disorders	31 (47.6%)
Personality disorders	11 (16.9%)
Generalized anxiety disorder	11 (16.9%)
Mood disorder (depression)	8 (12.3%)
Schizophrenia	1 (1.5%)

JME: juvenile myoclonic epilepsy; CAE: childhood absence epilepsy.

psychogenic non-epileptic seizures and two died from other causes than epilepsy. One patient without seizure control had sudden unexpected death after a follow-up period of seven years and his data were considered for analysis. Demographic and clinical data are presented in Table 1.

Patients had a follow-up of 5.72 ± 1.91 years. Control of each seizure type was verified as follows: myoclonia – good 53/65 (81.5%), moderate 4/65 (6.2%) and poor 8/65 (12.3%); GTCS – good 44/62 (71%), moderate 9/62 (14.5%) and poor 9/62 (14.5%); absences – good 21/31 (67.7%), moderate 4/31 (12.9%) and poor 6/31 (19.4%).

Forty of 65 (61.5%) patients reached good seizure control for all seizure types (Group 1), 25 (38.5%) of whom became completely seizure free. Twenty-five patients (38.5%), nine with moderate and 16 with poor seizure control constituted Group 2.

At the end of the study, four patients (6.2%) were seizure free and out of medications. The most common AED was VPA in 49/61 (80.3%), followed by TPM in 17 (27.9%), clonazepam (CZP) in 11 (18.0%), LTG in six (9.2%), PB in six (9.2%), clobazam in four (6.2%) and carbamazepine in two (3.1%). Thirty-eight of 61 (62.3%) medicated patients were on monotherapy: 31 (81.6%) on VPA, six (15.8%) on TPM and one (2.6%) on CZP.

Among 49 patients in use of VPA, 30 (61.2%) were in Group 1; 17 (34.7%) of them became seizure-free. Group 1 also included seven of 17 patients (41.1%) taking TPM, three (17.6%) of whom seizure-

Table 2
Prognosis related to clinical features.

	Group 1 (n = 40)	Group 2 (n = 25)	p-Value
Epilepsy duration (years)	8.7 ± 8.20	13.9 ± 9.0	0.019
Seizure types			
Myoclonia	2 (5.0%)	0	
Myoclonia + GTCS	25 (62.5%)	7 (28.0%)	
Myoclonia + GTCS + absences	12 (30.0%)	18 (72.0%)	0.003
Myoclonia + absences	1 (2.5%)	0	
JME subtypes			
Classic JME	33 (82.5%)	15 (60.0%)	
CAE/JME	4 (10.0%)	3 (12.0%)	
JME with adolescent onset pyknoleptic absence	3 (7.5%)	7 (28.0%)	0.067

JME: juvenile myoclonic epilepsy; CAE/JME: childhood absence epilepsy evolving to juvenile myoclonic epilepsy; GTCS: generalized tonic-clonic seizures; STAI: State-Trait Anxiety Inventory.

free. Five of six patients in use of PB were in Group 2 ($p = 0.038$) and had persistence of GTCS ($p = 0.041$). Five of six patients in use of LTG ($p = 0.038$) and 10 of 15 on benzodiazepines ($p = 0.033$) were also in Group 2. All nine patients on three or more AEDs belonged to the latter group ($p < 0.001$). Group 2 patients were in use of higher doses of VPA (1450 ± 879.24 mg vs. 811.7 ± 375.01 mg, $p = 0.001$) and of CZP (2.33 ± 1.03 vs. 0.80 ± 0.69 , $p = 0.02$); there was a higher number of Group 2 patients taking 2000–3500 mg of VPA daily ($p = 0.028$).

The analysis of clinical data revealed no differences between the two groups in relation to gender distribution (Group 1: 52.5% male vs. Group 2: 44%), family history of epilepsy (Group 1: 75% vs. Group 2: 64%) or follow-up duration (Group 1: 5.9 ± 1.93 vs. Group 2: 5.4 ± 1.87).

Table 2 shows correlation between clinical data and distribution of patients according to seizure control. Comparing to Group 1, patients in Group 2 had higher epilepsy duration and incidence of combination of all three seizure types, myoclonia, absences and GTCS. Regarding JME subtypes, Group 2 had a marginally significant higher representation of JME with adolescent onset pyknoleptic absence than the good seizure control group.

When we compared the 25 patients who became completely seizure-free with those with any degree of persistent seizures (Table 3), the latter presented earlier onset of epilepsy and higher incidence of cluster B personality disorders. While 87.5% of seizure-free patients had STAI slight degrees of anxiety traits, 68.8% of patients with persistent seizures had moderate to severe degrees of anxiety traits. In addition, patients with persistent seizures had significantly higher scores in anxiety-trait and marginally significant higher scores in anxiety-state subscale.

Table 3
Prognosis related to clinical features in seizure-free opposed to those with persistent seizures.

	Seizure-free patients (n = 25)	Patients with persistent seizures (n = 40)	p-Value
Age at epilepsy onset (years)	15.4 ± 5.47	12.6 ± 3.33	0.015
Personality disorders	1 (4%)	10 (25%)	0.029
Anxiety disorders			
Seizure-free patients (n = 16)		Patients with persistent seizures (n = 32)	p-Value
STAI (STATE) score	37.3 ± 12.0	44.1 ± 10.8	0.054
Slight degree	12 (75%)	14 (43.8%)	0.065
Moderate to severe degree	4 (25.0%)	18 (56.2%)	
STAI (TRACE) score	36.6 ± 11.4	45.9 ± 11.3	0.011
Slight degree	14 (87.5%)	10 (31.3%)	0.001
Moderate to severe degree	2 (12.5%)	22 (68.8%)	

STAI: State-Trait Anxiety Inventory.

On the other hand, patients with persistence of GTCS had higher duration (14.6 ± 8.35 vs. 8.9 ± 9.08 , $p = 0.028$) and earlier onset of epilepsy (12.1 ± 3.05 vs. 15.4 ± 5.47 , $p = 0.012$) than seizure-free patients.

3.2. Video-EEG

A higher number of patients with persistent seizures had epileptic discharges in baseline video-EEG (47.5% vs. 16%, $p = 0.016$) and also seizures recorded (55% vs. 12%, $p = 0.001$) in comparison with seizure-free patients.

3.3. Focal EEG abnormalities

Eleven patients had normal tracings. Among the remaining 54 patients, 27 (50%) had focal EEG abnormalities. Twenty-two had $\geq 50\%$ amplitude asymmetry of generalized discharges; 15 focal epileptiform discharges; four slow wave bursts; three focal onset of generalized discharges and two asymmetry of photoparoxysmal response. Among 17 patients whose EEG presented focal discharges or slow wave bursts, 10 had frontal localization, three parietal, three temporal and one, occipital.

Among 54 patients with epileptiform discharges, focal abnormalities were seen in 7/16 (43.8%) who became seizure-free and in 20 of 38 (52.6%) who had persistent seizures ($p = 0.76$).

3.4. Reflex traits

Fifteen of 27 (55.6%) patients with none or only general precipitants were seizure-free. For those with other modalities of precipitant factors the rate of seizure freedom dropped to 4/13 (30.8%) for eye-closure, 4/16 (25%) for photic, 5/17 (29.4%) for language and 5/22 (22.7%) for praxis sensitive patients.

Fig. 1 depicts prognosis related to reflex traits. In comparison to seizure-free, those with persistent seizures had a higher proportion of praxis (58.6% vs. 25%, $p = 0.04$) and language sensitivity (53.8% vs. 16.7%, $p = 0.026$). Photosensitivity was only marginally related to persistent seizures (50% vs. 21.1% in seizure-free patients, $p = 0.064$). Written calculation was considered among praxical activities. Mental calculation activated epileptic discharges in only four patients, none of them reached seizure freedom (Fig. 1). The prognosis related to seizures induction is in Fig. 2. All of the five patients with seizures triggered by IPS were not seizure-free (18.5% vs. 0.0%, $p = 0.05$); the same occurred with six patients with eye closure induced seizures (21.4% vs. 0.0%, $p = 0.025$). Also, praxis-induced seizures prevailed among non

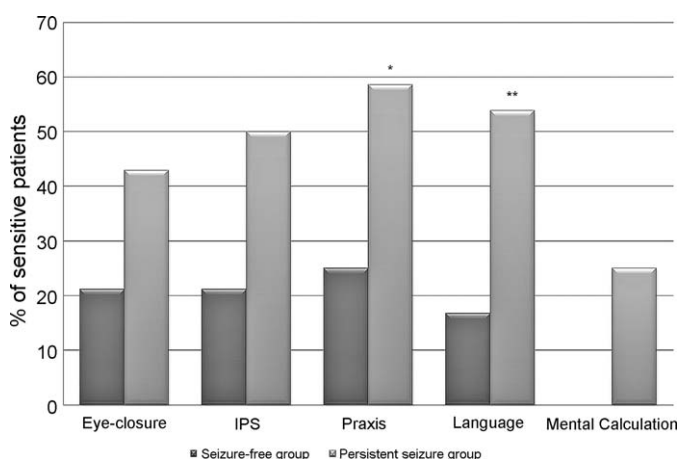


Fig. 1. Prognosis for patients with reflex traits expressed as EEG discharges and/or seizures induction. * $p = 0.04$; ** $p = 0.026$.

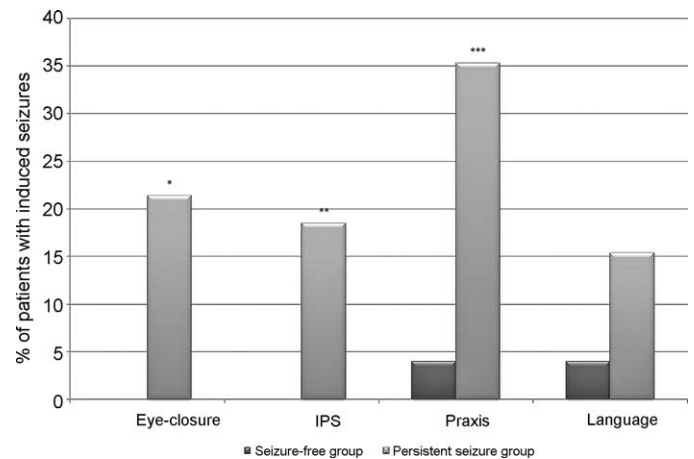


Fig. 2. Prognosis for patients with reflex traits expressed as induced seizures. IPS: intermittent photic stimulation. * $p = 0.025$; ** $p = 0.05$; *** $p = 0.004$.

seizure-free patients (35.3% vs. 4%, $p = 0.004$). There was no patient exhibiting seizures induced by mental calculation (Fig. 2).

Among 38 patients who had presented at least one reflex trait, 21 performed a second evaluation by EEG/video-EEG. The relation of reflex traits evolution and seizure control is represented in Fig. 3. Notice that most patients of Group 1 evolved with traits disappearance while most of those in Group 2 had traits persistence.

4. Discussion

In this prospective study we investigated the response to treatment in 65 patients with JME with three years of follow-up utilizing the seizure freedom criteria of Janz,²⁵ defined as a remission for at least two years. In Janz series of 49 JME patients, 25% did not become seizure free and 91% had seizure relapses after medication reduction or suspension.²⁵ Another study revealed unsatisfactory response to treatment in only four of 43 (9%) patients.²⁶ A prospective study with a follow-up of >3 years reported that 6/50 (12%) patients continued to have seizures.⁸ Recent studies have demonstrated variable rates of JME pharmacoresistance; in a population-based series of 13%¹² and in outpatient clinics of 15.5%⁹ and 30%.¹⁵ Responsiveness to

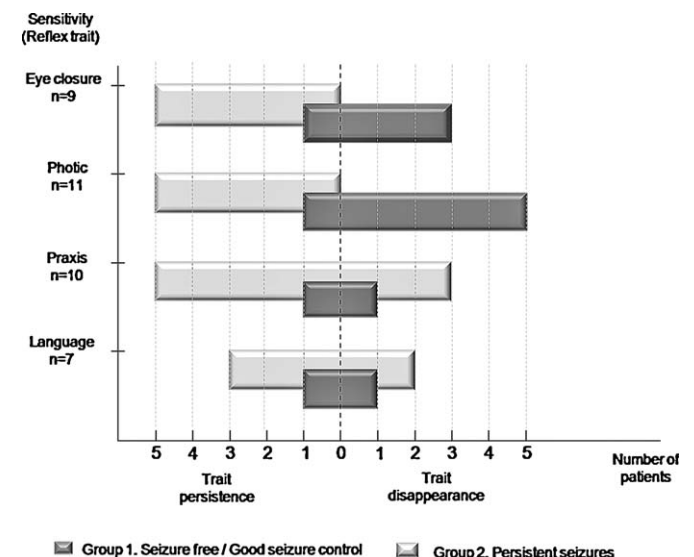


Fig. 3. Re-evaluation by EEG/video-EEG.

medication in JME has been defined in a variable way as completely seizure-free patients and those with persistence of occasional myoclonia or “rare” seizures, including GTCS or excluding this seizure type. Classification of patients regarding seizure control is not so clear when dealing with patients having some persistent seizures or relapsing ones. In this way, one study revealed persistence of GTCS in 18.6%, but that of other seizure types was not clearly quantified.²⁷ However, most of studies were retrospective, based on questionnaires or medical records review analyzing variable periods of follow-up from one to three or more years. Pseudo-resistance represented by the lack of compliance, insufficient treatment and abnormal lifestyle is another important factor in JME treatment, contributing to 9.7% of persisting seizures.⁹ Finally, the population studied was quite heterogeneous, varying from community-based to tertiary centers. It is believed that a higher proportion of refractory JME has been reported by specialized clinics probably due to the fact that difficult to treat patients are preferentially referred to such centers.²⁸ A more reliable evaluation of these patients in such institutions could also contribute to these results.

In our study performed in a tertiary university center we kept following 65 out of 76 (85.5%) patients for at least three years. The criteria for response to treatment were established for each seizure type and it was cautioned to exclude pseudo-refractory epilepsy patients. We found 25 out of 65 (38.5%) patients with moderate and poor seizure control. Up to now, the highest reported rate in patients followed for three years had been 45% for GTCS.¹⁴ Both samples might have been biased with representation of more severe epilepsy.

Despite the fact that there are no class I studies to guide initial therapy of JME,²⁹ according to SANAD, a large randomized, prospective trial, reporting on 119 patients with JME, VPA should remain as drug of choice for the majority of generalized and non-classified epilepsies, as it was more effective than LTG and better tolerated than TPM with the possible exception of childbearing age women.³⁰ Open-label studies using VPA in JME showed a 41–88% seizure-free rate for patients receiving this AED as monotherapy or add-on.^{9,31} In Nicolson et al. series,³² 54.3% of 962 IGE patients achieved one year period of remission; this was most likely with VPA monotherapy (52.1%), with lower rates for LTG and TPM (16.7% and 34.6%, respectively). Although the number of patients treated in our series was much smaller, we found seizure freedom outcome as 34.7% for VPA followed by 17.6% for TPM and 14.3% for LTG. Our data regarding response to specific AED suggest low efficacy of PB and LTG in JME since from the six patients receiving PB, only one became seizure-free, the same occurring to the six patients receiving LTG.

The present study did not have as its main target to estimate the rate of non-responders but to look for features that could differentiate the group with refractory or more difficult to control epilepsy.

In this series, seizure persistence was related to higher duration of epilepsy, similar to other reports.^{14,33} Otherwise, patients who did not become seizure-free had a lower age at epilepsy onset. Earlier onset of epilepsy was also emphasized in the resistant form of childhood absence epilepsy and might be an expression of different genetic basis.³⁴

Regarding seizure types, myoclonic seizures alone, in combination to GTCS on awakening or to absences were related to a better prognosis when compared to the combination of all the three seizure types in the same patient,^{9,10,14} a fact also observed in the present series.

Different subtypes of JME patients defined by clinical and EEG criteria had no difference regarding seizure control and it was observed only a tendency toward a worse seizure control in the group with adolescent onset pyknoleptic absence. This is different

from the original description, in which this subtype accounted for similar non-responsive rates as classical subtype (44 and 42%, respectively) while childhood absence epilepsy evolving to JME (subtype 2) had 91.4% of patients not seizure-free.¹³

Regarding psychiatric disorders, previous reports of higher frequency of personality disorders and generalized anxiety disorder in patients with drug resistance^{9,17} were confirmed in this study. Higher degrees of anxiety measured by trace STAI scale were found among no seizure-free patients. Psychiatric disorders might share pathophysiological mechanisms with JME, both related to dysfunction in fronto-thalamic circuitry or the psychiatric disturbance *per se* might have a negative impact over epilepsy treatment response. In fact, current evidence exists of more severe abnormalities demonstrated by spectroscopy and voxel-based volumetry in thalamic regions and frontal lobes, in JME patients with psychiatric disorders compared to those without such finding.^{35,36}

As to EEG features, among the characteristics associated to persistence of seizures, presence of epileptic discharges in baseline and seizure registration might suggest these patients had a higher epilepsy severity since the beginning of the follow-up period.

The articles dealing with focal EEG abnormalities in JME do considerations concerning the difficulties they add to diagnosis and advocate controversial implications of this feature in prognosis. While some authors^{37,38} described no correlation between focal abnormalities and response to treatment, others emphasized they should not be misinterpreted as indicative of focal epilepsies, a common mistake in JME, and its importance as a sign of worse prognosis.^{15,16} Our analysis on the presence of focal EEG abnormalities did not reveal a significant difference regarding seizure control, although some possible associations might be suggested. In this manner, two of three patients with parietal and six of 10 with frontal abnormalities had praxis induction. In praxis sensitivity, the most affected region may be in either the frontal or the parietal lobes, as suggested by Goossens et al.³⁹ The patient with occipital abnormalities was photosensitive and all of three with temporal abnormalities had psychiatric disorders, two with generalized anxiety disorder and one, cluster B personality disorder.

Reflex traits might have implication on prognosis. Matsuoka¹⁴ for the first time reported the negative impact of neuropsychological activation on prognosis of JME. Inoue and Kubota⁶ analyzing a series of 134 JME patients reported that seizure-freedom for more than three years dropped from 53/77 (69%) in patients with none or only general precipitants, to 18/32 (56%) in photosensitive and 12/25 (48%) for praxis sensitive patients. This, therefore, suggested that the presence of reflex traits could represent an aggravating feature in JME. In this sense, our study showed that the rate of seizure freedom dropped from 15/27 (55.6%) in patients with none or general precipitant factors to 4/16 (25%) in photosensitive and to 5/22 (22.7%) for praxis sensitive patients, the latter reaching statistical significance level.

Approximately 30% of JME patients have a photoparoxysmal EEG response, although most articles do not refer to prognostic significance of this trait in JME.⁴⁰ A very good response to VPA has been reported among photosensitive patients, especially in pure photosensitive epilepsy and in JME,^{38,40,41} although no single – or double-blind trials to evaluate treatments in visually induced seizures have been performed and even VPA, the drug of choice has never been investigated as such.⁴² In our series eye-closure and photosensitivity were related to a worse prognosis when inducing clinical seizures. Additionally while only one out of the six Group 1 patients continued expressing the trait, all in Group 2 were still photosensitive at the end of the study. Another series examined 50 patients with photosensitive epilepsies and found that abnormalities in IPS occurred in 76% of patients with recent seizures vs. in 33% of seizure-free patients for more than one year.⁴³

Similarly, in the present study language sensitivity had a negative impact on prognosis. Although reading epilepsy has been described as having a good prognosis,^{44,45} this might not be true in language-induced seizures in JME. In this respect, among nine patients in Mayer et al. series,⁵ three were not seizure-free on VPA as 14/17 (82.3%) of language sensitive patients in ours.

Calculation activated epileptiform discharges in four patients and at the end all were not seizure-free. Seizures induced by calculation using a “soroban” (Japanese abacus), could not be induced by mental calculation.⁴⁶ The latter would exert a higher provocative effect when associated to praxis as in the use of soroban or written calculation. In fact, for the majority of JME patients, mental calculation might have an inhibitory effect presumably by the activation of circuitries that inhibits activity in the motor frontal area more related to ictogenesis.²³

Reflex traits evolution over time was described in a study on two JME patients evaluated for more than 20 years whose susceptibility to seizures during performance of praxis tasks lasted all period, although the effect decreased with age.¹⁹ The repeated exams in our study suggested a parallel evolution of seizure control and reflex traits expression, with persistent traits in the majority of Group 2 patients and disappearance in most of Group 1.

Data on reflex traits indicate they are expression of hyperexcitability in JME patients. The most recent theory on IGE mechanisms attributes a fundamental role of cortex in initiating spike and wave discharges.⁴⁷ Hyperexcitability in JME probably is not homogeneous in the entire cortex and reflex traits might be the expression of higher excitability in specific cortical regions.

5. Conclusions

Clinical features including reflex traits have prognosis implications in JME. Longer epilepsy duration, combination of all three seizure types, discharges in baseline video-EEG, seizures recording and praxis sensitivity were negative prognostic factors for seizure control. Additionally, a lower rate of seizure freedom was associated to earlier age at epilepsy onset, personality disorders, anxiety, sensitivity to praxis or to language tasks and seizure induction by eye-closure or IPS. Reflex traits disappearance was generally accompanied by seizure control. All these endophenotypes might serve as prognosis signaling in JME.

Conflict of interest

None of the authors has any conflict of interest to disclose.

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